

 VIT-AP UNIVERSITY	Continuous Assessment Test – Long Summer (2024-25) - June 2025	
	Maximum Marks: 50	Duration: 90 Mins
Course Code: CSE2008	Course Title: Operating Systems	
Set No: 1	Exam Type: Closed Book	School: SCOPE
Date: 16/06/2025	Slot: A	Session: AN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any Open Book/Open Notebook/Closed Book:		
1. "fx series" - non Programmable calculator are permitted : YES 2. Reference tables permitted : NO		

PART – A: Answer any ALL Questions, Each Question Carries 10 Marks (5×10=50 Marks)

1. write a C program opens a file, reads its contents, displays them on the screen, and then exits. Write the sequence of system calls that the operating system must process, and classify each into its respective system call type. (10 M)
2. An IT professional compares early batch systems to modern operating systems and claims that batch systems are obsolete. What features have evolved in modern systems to make them more versatile? Discuss your answer by relating it to the different types of operating systems. (10 M)
3. Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. In answering the questions, use nonpreemptive scheduling, and base all decisions on the information you have at the time the decision must be made. (10 M)

Process	Arrival Time	Burst Time
P1	0.0	8
P2	0.4	4
P3	1.0	1

- a. What is the average turnaround time for these processes with the SJF scheduling algorithm?
 - b. The SJF algorithm is supposed to improve performance, but notice that we chose to run the process *P1* at time 0 because we did not know that two shorter processes would arrive soon. Compute what the average turnaround time will be if the CPU is left idle for the first 1 unit and then SJF scheduling is used. Remember that processes *P1* and *P2* are waiting during this idle time, so their waiting time may increase. This algorithm could be known as *future-knowledge scheduling*.
4. Explain the different states of a process in an operating system. Describe each state clearly and illustrate the possible transitions between them using a neat state transition diagram. (10 M)
 5. What is a semaphore in operating systems? Explain the two types of semaphores (binary and counting) and show how semaphores can be used to solve the critical-section problem with a producer-consumer example. (10 M)

QP MAPPING

Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	E	1	10	3	CO1	PO1, PO2		
Q2	A	1	10	4	CO2	PO1, PO3		
Q3	A	2	10	4	CO4	PO1, PO2, PO5		
Q4	A	2	10	2	CO2	PO1		
Q5	T	3	10	3	CO3	PO1, PO2		